Day 101 | Unit 07 | Teacher plan

Understand and Use Percent

Find the Unknown Percent

Standards: 6.RP.A.3c

Learning target: I can find the percent that one quantity is of another.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

70 is what percent of 140?

1. Divide the part by the whole, then multiply by 100.

2. Keep the same units in part and whole.

3. 70 ÷ 140 × 100 = 50%

4. the reference whole is 140.

5. State the result in the form requested: 50%.

Answer: 50%

Worked example 2

A quantity changes from a reference whole of 60 to a compared amount of 45. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (45 ÷ 60) × 100 = 75%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 75%; denominator 60

Day 101 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

40 is what percent of 200?

02 / Constructed response

12 is what percent of 120?

03 / Constructed response

80 is what percent of 160?

04 / Constructed response

125 is what percent of 100?

05 / Constructed response

75 is what percent of 100?

06 / Constructed response

40 is what percent of 100?

07 / Constructed response

30 is what percent of 40? State which quantity is the reference whole.

08 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 15. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

09 / Numeric entry and explanation

A quantity changes from a reference whole of 60 to a compared amount of 36. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

24 is what percent of 60? State which quantity is the reference whole.

Day 101 | Unit 07 | Answer key

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Learning target: I can find the percent that one quantity is of another.

1. 20%

40 ÷ 200 × 100 = 20%; the reference whole is 200.

6.RP.A.3c | Constructed response

2. 10%

12 ÷ 120 × 100 = 10%; the reference whole is 120.

6.RP.A.3c | Constructed response

3. 50%

80 ÷ 160 × 100 = 50%; the reference whole is 160.

6.RP.A.3c | Constructed response

4. 125%

125 ÷ 100 × 100 = 125%; the reference whole is 100.

6.RP.A.3c | Constructed response

5. 75%

75 ÷ 100 × 100 = 75%; the reference whole is 100.

6.RP.A.3c | Constructed response

6. 40%

40 ÷ 100 × 100 = 40%; the reference whole is 100.

6.RP.A.3c | Constructed response

7. 75%

30 ÷ 40 × 100 = 75%; the reference whole is 40.

6.RP.A.3c | Constructed response

8. 15%; denominator 100

Percent = (15 ÷ 100) × 100 = 15%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

9. 60%; denominator 60

Percent = (36 ÷ 60) × 100 = 60%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 40%

24 ÷ 60 × 100 = 40%; the reference whole is 60.

6.RP.A.3c | Constructed response